

Tingjia Xu

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PROFILE

Research Interests: • Active Mobility • Built Environment & Travel Behavior • Healthy Cities • Urban Design

Selected Highlights: • World Bank Short-Term Consultant • Route Choice Analysis based on 3M+ Cycling Trajectories
• Project Lead for a Healthy Community Regeneration Handbook

EDUCATION

Tongji University, Shanghai, China

Sep. 2023–Jun. 2026

M.Eng. in Urban Planning | GPA: 4.54/5.00 (Top 2%) | Healthy City WLAN Lab | Advisor: Prof. Lan Wang

- China National Scholarship (2024, ranked 1st in cohort)
- Shanghai Outstanding Graduate Award (2026); Tongji University Outstanding Student (two consecutive years)

Tongji University, Shanghai, China

Sep. 2018–Jun. 2023

B.Eng. in Urban and Rural Planning | GPA: 4.40/5.00 (Top 3%)

- Tongji University Outstanding Undergraduate Thesis
- First-Class Scholarship; College Dean's Scholarship (highest college-level award)

SELECTED RESEARCH EXPERIENCE

The Impact of Built Environment on Cycling Route Choice: Evidence and Planning

Sep. 2023–Jun. 2026

Implications from Shanghai Using Mobike Trajectory Data

Master's Thesis | Advisor: Prof. Lan Wang | Chinese review under review | English empirical manuscript in final revision

- Map-matched 3M+ dockless bike-sharing trajectories and extracted streetscape features using semantic segmentation.
- Leveraged GPS data to derive choice sets, improving behavioral plausibility while reducing algorithmic reliance.
- Estimated Path Size Logit models to identify time-of-day variation in cyclists' route preferences across the morning peak, off-peak daytime, evening peak, and nighttime periods, informing bicycle-friendly street improvements.

Emergency Medical Services (EMS) Risk Assessment and Decision Optimization Platform

May 2026–Present

Student-led Research Commercialization Project | Main Researcher | Healthy City WLAN Lab | Project Lead: Dr. Surong Zhang

- Developed and backtested a spatiotemporal model for emergency-call demand prediction.
- Built the web platform and UI/UX, integrating demand forecasts with interactive map-based visualization.
- Supported a commercialization partnership with a leading EMS dispatch provider serving 80% of the national market.

Quantifying Heat-Adjusted Health Benefits of Cycling in Cities

Jun. 2025–Present

National Key R&D Program of China | Main Researcher | Advisors: Prof. Lan Wang; Asst. Prof. Wenyao Sun

- Contributed to the development of the Heat-Exposure-Adjusted Cycling Health Benefit Index (HCBI).
- Processed multi-source data to estimate HCBI at 100-m resolution across Shanghai.
- Co-authored an equal-contribution working paper and drafted a U.S. patent application (pending).

Multi-Objective Streetscape Planning for Thermal Environment and Cycling Vitality

Jun. 2025–Jun. 2026

National Key R&D Program of China | Graduate Researcher | Advisors: Prof. Lan Wang; Asst. Prof. Lu Shan

- Built a street-segment dataset integrating cycling volume, thermal conditions, and built-environment attributes.
- Conducted the literature review for a second-author Cities manuscript (under revision).

Social Media Sentiment Analysis for Perceived Livability

Apr. 2025–Nov. 2025

National Key R&D Program of China | Graduate Researcher | Advisors: Prof. Lan Wang; Assoc. Prof. Bochu Liu

- Developed a four-dimensional livability lexicon to identify relevant social media posts for sentiment analysis.
- Reviewed the literature on walkability and bikeability assessment dimensions and spatial indicators.

Space Syntax Analysis of Urban Underground Space

Oct. 2024–Jan. 2025

Graduate Course Project | Team Lead | Advisor: Prof. Yu Ye | Presented at the 65th ACSP Annual Conference

- Used Space Syntax to assess accessibility and intelligibility at Jing'an Temple Station.
- Applied visibility graph analysis and agent-based movement simulation to optimize retail placement for pedestrian exposure and commercial potential.

Skills: Python · ArcGIS · Map matching · Discrete choice modeling · Street-view semantic segmentation · Space syntax · sDNA

PUBLICATIONS

Manuscripts Under Review

- [1] **Xu, T.**, Wang, L.* , Zhang, S., & Sun, W. Research progress and future prospects on the impact of street space factors on cycling route choice. *Urban Development Studies* (under review; in Chinese).
- [2] Shan, L., **Xu, T.**, Zhou, Y., Zhang, S., & Liu, B. AI-enhanced livable environment planning: Streetscape synergies on urban thermal environment and cycling vitality in Shanghai. *Cities* (under revision).
- [3] Sun, W., Wang, L.* , **Xu, T.**, Wang, Y., Kwan, M.-P., Yang, X., Zhang, Y., Zhang, S., Chen, L., Li, Q., & Wang, C. Balancing total health gains and equity: Evaluating cooling effects of resource-constrained urban greening strategies. *Cities* (under revision).

Peer-Reviewed Publications

- [4] Zhang, S., Wang, L.* , Shen, Y., Zhang, Y., Gao, Y., **Xu, T.**, & Zhang, Z. (2024). What causes Emergency Medical Services (EMS) delay? Unravel high-risk buildings using citywide ambulance trajectory data. *Habitat International*, 153, 103198. doi: 10.1016/j.habitatint.2024.103198

Working Papers

- [5] **Xu, T.**, Wang, L. *, Zhang, S., & Jia, Y. How do cyclists' street-environment preferences vary within a day? Evidence from dockless bike-sharing route choices in Shanghai (In final revision for submission).
- [6] Sun, W.†, **Xu, T.**†, Wang, L.* , Kwan, M.-P., Wang, Y., & Li, X. Quantifying heat-adjusted health benefits of cycling in cities.

CONFERENCE PRESENTATIONS

- Xu, T.**, Wang, L., Zhao, Y., & Tao, J. (2025). Rethinking underground space utilization in metro stations: A case study of Shanghai's Jing'an Temple. 65th ACSP Annual Conference, Minneapolis, MN, USA.
- Xu, T.**, Wang, L., & Zhang, S. (2025). Insights for cycle-friendly streets: Unveiling dynamic cycling route preferences. 19th IACP Conference, Xiamen, China.
- Xu, T.**, Yang, H., Ben, B., & Zhang, Y. (2025). Reconstructing the healing system of public space: Data-driven health interventions and inclusive practices in high-density urban areas. 37th AESOP Annual Congress, Istanbul, Türkiye (virtual presentation).

WORK EXPERIENCE

World Bank Group, Office of the Chief Economist for Infrastructure (Washington, DC) May 2026–Present
Short-Term Consultant / China (remote) / Project Lead: Yue Li, Senior Economist

- Developing a geospatial database of 300+ large-scale urban development projects in China and processing project data for Indonesia.

Asian Development Bank (ADB)–Tongji Urban Knowledge Hub | Intern, Shanghai Sep. 2023–Jun. 2024

- Drafted urban development reports based on international case reviews (e.g., a report on health-promoting community regeneration).

Shanghai Tongji Urban Planning & Design Institute | Intern, Shanghai Sep. 2023–Dec. 2025

- Conducted fieldwork, GIS analysis, and urban design across four planning projects (e.g., 15-minute community life-circle planning).

INNOVATION PROJECTS & PRACTICE

Design Handbook for Health-Promoting Regeneration of Community Public Spaces Project Lead
National Innovation Project / 2022–2023

- Developed tailored handbook versions for planners and residents and established a WeChat feedback channel.

A Complementary Emergency-Response Platform for Emergency Medical Services (EMS) Deputy Lead
National Bronze Award, China International College Students' Innovation Competition / 2024

- Led the final presentation and contributed to translating research into an implementable EMS response proposal.

Other Selected Leadership

- CAMPUS Asia–SUAE Winter School, Pusan National University, South Korea (2024) Team Lead
- University Key Project | Urban AED Deployment Assessment and Planning Optimization (2024) Project Lead
- Career Development Center, Tongji CAUP Graduate Student Union (2023–2024) Director

SELECTED COMPETITION AWARDS

Awards in planning, design & innovation: 4 national, 3 provincial, and 6 university-level.

2023 Future Planners Graduate Planning Design & Thesis Competition	National	First Prize (1/1)
2024 Future Planners National Spatial Planning Design Competition	National	First Prize (2/5)
2024 Beautiful China Graduate Innovation Design Competition	National	First Prize (2/5)
2024 Shanghai “Huichuang Youth” Urban Design Competition	Provincial	First Prize (1/1)